

A. 3-gram Feature Contribution Experiment

In the above ablation experiments, the results demonstrate that the joint modeling of 2-gram and 3-gram character features provides effective semantic support for the entity alignment task. To further evaluate the specific contribution of 3-gram features to model performance, we conduct comparative experiments on the three sub-datasets of DBP15k (ZH-EN, JA-EN, FR-EN). We remove 3-gram character features while keeping all other model structures and training strategies unchanged, retaining only 2-gram features for comparison. The experimental results are shown in Table 1

Table 1 PERFORMANCE DEGRADATION AFTER REMOVING 3-GRAM FEATURES

	CLEFA- EA-L Hits@1	CLEFA- EA-S Hits@1	CLEFA- EA-S Hits@10	CLEFA- EA-S MRR
DBP15k _{zh-en}	93.06	91.82	97.07	0.94
DBP15k _{ja-en}	96.52	95.31	98.91	0.96
DBP15k _{fr-en}	99.60	98.93	99.85	0.99
Average Drop (vs.2+3gram)	0.83	0.47	0.42	0.01

It can be seen from the results that the model performance decreases to a certain extent on all three sub-datasets after removing 3-gram features. Specifically, CLEFA-EA-L, which focuses on hard alignment accuracy, achieves an average performance drop of 0.83 in Hits@1. CLEFA-EA-S, which focuses on soft alignment, shows an average decrease of about 0.4 in both Hits@1 and Hits@10, while the MRR metric changes slightly. These results indicate that 3-gram character features can supplement the sub-word information captured by 2-gram to a certain extent, providing finer-grained semantic discrimination for entity alignment. Nevertheless, its overall contribution is relatively limited, and the model remains robust to 2-gram features.

In addition, removing 3-gram features has the least impact on model performance on the DBP15k_{fr-en} sub-dataset. This is likely because French and English share high consistency in alphabet, word structure, and entity naming conventions, so 2-gram character features can already effectively characterize local similarities between cross-lingual entity names, resulting in relatively limited gains from 3-gram features. Compared with DBP15k_{ja-en} and DBP15k_{fr-en}, DBP15k_{zh-en} shows a more significant performance drop after removing 3-gram features. This is mainly due to the substantial differences between Chinese and English in writing system and word formation, with almost no shared character patterns. In this scenario, higher-order character n-grams (e.g., 3-gram) can capture relatively stable continuous character segments, providing critical alignment clues for cross-lingual entity names. Removing 3-gram features weakens such fine-grained sub-word information, thus exerting a more significant negative impact on alignment performance.

Theoretically, introducing even higher-order character n-gram features (e.g., 4-gram) may further enhance the model’s representation capacity. However, this will lead to an exponential growth in feature space size, significantly increasing computational and memory overheads. Observations on the DBP15k_{fr-en} dataset show that when combining 2-gram and 3-gram features, the corresponding feature dimensions reach 1622 and 9726 respectively, showing an obvious expansion trend. After further introducing 4-gram features, the character feature dimension surges to 41718, causing out-of-memory errors and unstable training in our experimental environment. Combined with the relatively limited performance gain brought by 3-gram features in the aforementioned experiments, we conclude that adding 4-gram features cannot achieve a reasonable balance between performance improvement and computational cost. Therefore, considering the model’s lightweight design goal and experimental feasibility, this paper does not include 4-gram features in the final experimental setup, and adopts the combination of 2-gram and 3-gram as the character-level modeling scheme.

B. Graph Propagation Layer Selection Experiment

The propagation layer number L is a critical parameter in the graph propagation mechanism, and its value directly affects the model’s ability to model multi-hop structural semantic information. Although existing research (COEA) has shown through experiments that fixing L to 2 can achieve the best performance, the optimal value of this parameter still needs to be re-evaluated

after introducing different feature interactions and alignment strategies. Based on this, this paper conducts comparative experiments on different propagation layers on the DBP15k dataset.

The experimental results are as follows. Table 2 shows the CLEFA-EA-L Hits@1 values on the three sub-datasets of DBP15k. Table 3 and Table 4 show the CLEFA-EA-S Hits@1 values and Hits@10 values on the three sub-datasets of DBP15k respectively. Figure 1 shows the combined line charts of the above experimental results, for a more intuitive illustration of optimal parameter selection. Since MRR remains almost unchanged, it is not listed.

Table 2 CLEFA-EA-L HITS@1 VALUES UNDER DIFFERENT L VALUES

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
L=1	93.46	95.40	99.22
L=2	93.84	96.90	99.68
L=3	95.59	97.88	99.68
L=4	95.64	97.92	99.68
L=5	95.64	97.86	99.60

Table 3 CLEFA-EA-S HITS@1 VALUES UNDER DIFFERENT L VALUES

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
L=1	92.30	93.93	98.34
L=2	93.47	95.38	98.62
L=3	93.71	95.63	98.49
L=4	93.45	95.29	98.31
L=5	92.94	94.63	98.00

Table 4 CLEFA-EA-S HITS@10 VALUES UNDER DIFFERENT L VALUES

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
L=1	97.18	98.44	99.76
L=2	98.04	99.16	99.88
L=3	98.33	99.21	99.92
L=4	98.30	99.19	99.87
L=5	98.19	99.11	99.84

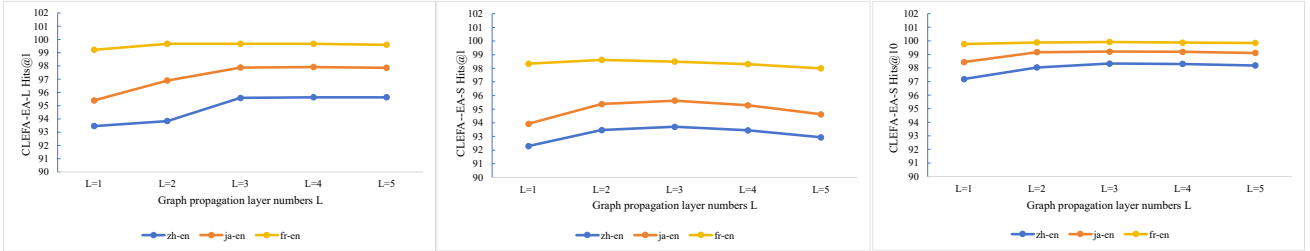


Figure 1 Performance trends of CLEFA-EA under different graph propagation layer numbers

As shown in Table 2 to Table 4, the experimental results indicate that with the increase of L, the model performance on the three sub-datasets of DBP15k generally shows a gradual improvement trend from shallow to deep, and reaches the optimal or sub-optimal performance around L=3.

Specifically, under the CLEFA-EA-L setting, when L increases from 1 to 3, the Hits@1 of the model on the DBP15k_{zh-en} and DBP15k_{ja-en} datasets improves most significantly; when $L \geq 3$, the performance gain tends to saturate, and continuing to increase the number of propagation layers only brings a very small improvement. This indicates that three-layer graph propagation has been able to fully capture the structural semantic information required for cross-knowledge graph alignment.

Under the CLEFA-EA-S setting, both Hits@1 and Hits@10 reach the peak or near-peak at L=3, while the performance decreases to varying degrees when L>3. This indicates that excessively deep propagation may introduce noise or lead to the over-smoothing problem of representations. Therefore, L=3 achieves a better balance between expressive ability and noise suppression.

Considering different settings and evaluation indicators comprehensively, this paper finally selects L=3 as the default parameter setting for the number of graph propagation layers. This choice ensures the model performance while avoiding unnecessary computational overhead, and shows good stability on multiple datasets.

C. Training Epoch Sensitivity Experiment

As an entity alignment method based on contrastive learning, the model needs to gradually pull positive samples and separate negative samples through multiple iterations during training. Its performance may fluctuate with the number of training epochs. If the model is highly sensitive to training epochs, it may lead to insufficient stability under different experimental settings. Based on this, this paper further analyzes the impact of training epochs on the performance of the CLEFA-EA model, and evaluates the alignment performance of the model on the DBP15k dataset under different numbers of training epochs to verify the convergence characteristics and robustness of the model. The experimental results are shown as follows.

Table 5 Hits@1 VALUES OF CLEFA-EA-L UNDER DIFFERENT TRAINING EPOCHS

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
Epoch=50	95.60	97.82	99.67
Epoch=100	95.60	97.88	99.68
Epoch=150	95.59	97.88	99.68
Epoch=200	95.59	97.88	99.68

Table 6 Hits@1 VALUES OF CLEFA-EA-S UNDER DIFFERENT TRAINING EPOCHS

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
Epoch=50	93.69	95.60	98.48
Epoch=100	93.71	95.60	98.48
Epoch=150	93.72	95.62	98.48
Epoch=200	93.71	95.63	98.49

Table 7 Hits@10 VALUES OF CLEFA-EA-S UNDER DIFFERENT TRAINING EPOCHS

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
Epoch=50	98.31	99.21	99.91
Epoch=100	98.32	99.21	99.92
Epoch=150	98.33	99.21	99.92
Epoch=200	98.33	99.21	99.92

Table 5 – Table 7 show the entity alignment performance of CLEFA-EA on the three cross-lingual sub-datasets of DBP15k under different numbers of training epochs. It can be observed that the model performance shows a trend of rapid convergence first and then stabilization with the increase of training epochs, regardless of the strict one-to-one alignment setting based on LAPJV (CLEFA-EA-L) or the soft alignment setting based on Sinkhorn (CLEFA-EA-S).

Specifically, under the CLEFA-EA-L setting, when the number of training epochs reaches 50, the Hits@1 metric of the model on the three sub-datasets is close to the optimal level. Continuing to increase the number of training epochs to 100 or more brings extremely limited performance improvement, and the value almost remains unchanged after 150 epochs. This indicates that the model can achieve effective convergence within a relatively small number of training epochs, and can obtain stable and high-quality entity representations without relying on long-term training.

Under the CLEFA-EA-S setting, the fluctuation ranges of Hits@1 and Hits@10 metrics under different numbers of training epochs are also small, and the performance variation on each dataset is controlled within 0.1%, further verifying the stability of the model in the soft alignment scenario. This phenomenon shows that contrastive learning and hard negative sampling mechanism can effectively guide the structured distribution of the entity feature space, enabling the model to form a relatively stable alignment discrimination ability in the early training stage.

To sum up, the CLEFA-EA model is not sensitive to the number of training epochs and has good convergence characteristics and robustness. On the premise of ensuring alignment accuracy, the model can be trained in a relatively small number of epochs, which helps to reduce the overall training cost and improve the scalability and practical value of the model in real application scenarios.

D. Temperature Coefficient Sensitivity Experiment

In the contrastive learning-based entity alignment framework, the temperature coefficient τ is used to adjust the smoothness of the similarity distribution, thereby controlling the discrimination intensity of positive and negative sample pairs in the loss function. Different values of τ may affect the convergence behavior and final alignment performance of the model. To evaluate the sensitivity of the CLEFA-EA model to the setting of the temperature coefficient, this paper conducts a comparative analysis on the Hits@1 metric under different values of τ with the model structure and other training parameters fixed, in both CLEFA-EA-L and CLEFA-EA-S settings. The experimental results are shown in 0 and 0

Table 8 Hits@1 VALUES OF CLEFA-EA-L UNDER DIFFERENT TEMPERATURE COEFFICIENTS

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
$\tau=0.01$	94.73	97.14	99.59
$\tau=0.03$	95.29	97.50	99.66
$\tau=0.05$	95.60	97.88	99.68
$\tau=0.07$	95.80	97.57	99.68
$\tau=0.1$	95.71	97.58	99.69

Table 9 Hits@1 VALUES OF CLEFA-EA-S UNDER DIFFERENT TEMPERATURE COEFFICIENTS

	DBP15k _{zh-en}	DBP15k _{ja-en}	DBP15k _{fr-en}
$\tau=0.01$	92.76	94.57	98.18
$\tau=0.03$	93.27	95.14	98.30
$\tau=0.05$	93.71	95.63	98.49
$\tau=0.07$	93.81	95.71	98.58
$\tau=0.1$	94.04	95.67	98.62

Table 8 and Table 9 present the Hits@1 alignment performance of CLEFA-EA under CLEFA-EA-L and CLEFA-EA-S settings with different temperature coefficients τ , respectively. Overall, the model achieves stable alignment performance over a wide range of τ values without severe performance fluctuations, indicating that CLEFA-EA is robust to the setting of the temperature coefficient.

Under the CLEFA-EA-L setting, when τ increases from 0.01 to 0.05, the Hits@1 metric of the model on all three sub-datasets shows a clear improvement. When τ is further increased to 0.07 and 0.1, the overall model performance remains stable with only minor fluctuations across datasets. Specifically, the best Hits@1 performance on the DBP15k_{zh-en} dataset is achieved at $\tau = 0.07$, while the performance differences among different τ values are negligible on the DBP15k_{fr-en} dataset, suggesting that the model is insensitive to changes in the temperature coefficient in high-precision alignment scenarios.

Under the CLEFA-EA-S setting, the model shows a steady upward trend with the increase of τ and achieves optimal or near-optimal alignment performance at $\tau = 0.07$ and $\tau = 0.1$. This phenomenon indicates that in soft alignment scenarios, a slightly larger temperature coefficient helps alleviate gradient fluctuations caused by contrastive learning while maintaining discriminability, thereby improving overall alignment performance.

Based on the above experimental results, CLEFA-EA can achieve stable and favorable alignment performance when $\tau \in [0.05, 0.1]$, and the model does not rely on a specific value of the temperature coefficient. Considering both performance and stability, this paper adopts $\tau = 0.05$ as the default setting in the experiments.